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When to go electric? A parallel bus fleet replacement study

Adrita Islam[a,*], Nicholas Lownes[b]

[a] Fehr & Peers, Sacramento, CA 95833, United States
[b] Department of Civil and Environmental Engineering, University of Connecticut, Storrs, CT 06269, United States

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ABSTRACT

United States in 2017 emitted about 14.36% of the total global Greenhouse Gas (GHG), 27% of which comes from the transportation sector. In order to address some of these emission sources, alternative fuel technology vehicles are becoming more progressive and market ready. Transit agencies are making an effort to reduce their carbon footprint by adopting these technologies. The overarching objective of this paper is to aid transit agencies make more informed decisions regarding the process of replacing a diesel fleet with alternative-technology buses to minimize GHG emissions. This study investigates the complete course of fleet replacement using a deterministic mixed integer programming. Bus fleet replacement is optimized by minimizing the Life Cycle Cost (LCC) of owning and operating a fleet of buses and required infrastructures while reducing GHG emission simultaneously. Buses operated by Connecticut Department of Transportation (CTDOT) were used as a case-study. Results also show a significant reduction in both cost and emission for optimized replacement schedule vs the unoptimized one. Results also show that a fleet consisting of 79% Battery Electric Bus and 21% Diesel Hybrid Bus yields the least cost solution which conforms to the other operational and environmental constraints. This study also includes various sensitivity tests, that illustrates that although the magnitude of the results may vary depending on the input data, the direction remains the same. The problem formulated in this study can help any transit agencies determine the most optimized solution to their fleet replacement problem under customizable constraints or desired set of outcomes.

1. Introduction

The increase in carbon dioxide (CO_2) emissions has been of broad and current interest for several decades around the globe. Global Carbon Dioxide emission, which is the main constituent of greenhouse gas (GHG), has increased about 90% since 1970, of which combustion of fossil fuel contributed about 78% (Boden et al., 2017). Total 2014 greenhouse gas (GHG) emissions in the U.S. was 6793.69 million metric tons CO_2 equivalent (MMT CO_2 e). Combustion of fossil fuels for transportation purposes is the second largest contributor of GHG emission in the U.S., contributing about 27% of the total GHG (EPA, 2017a, 2017b). In the state of Connecticut, it is about 36% of total GHG emissions (EIA, 2018). Although transit buses account for less than 1% of the total transportation emissions, they are significant sources of several pollutants that lead to ground level ozone formation and smog and eventually lead to public health hazards (EPA, 1990). While the average single occupant auto emits 0.44 kg of CO_2 per passenger mile (PPM), the average public transit bus emitted only 0.29 kg CO_2PPM per bus mile; with all seats occupied, it would emit only 0.08 kg CO_2 PPM (Weigel et al., 2009). But, if a bus operates regularly with a lower average occupancy, the emission associated with it becomes much higher. So, investment in low-carbon and low pollutant mobility technology and increasing the market share of public

* Corresponding author.
E-mail address: a.islam@fehrandpeers.com (A. Islam).

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transport should be given equal significance.

Diesel combustion engines are still the predominant type of fuel technology for transit buses in the United States. Compressed natural gas (CNG), Liquefied Natural Gas (LNG), and Biodiesel are examples of alternate fuels gases which have lower emissions, while hybrid diesel-electric buses that combine internal combustion engine propulsion with an electric is becoming increasingly popular as well. Although they use diesel as their primary fuel, these buses are more fuel efficient and are considered "green" as they emit less GHG. In 2015, 46.9% of U.S. public transportation buses were using alternate fuel or hybrid technology (APTA, 2015) which is a significant change compared to that of 2010 being 33.5% and 2005 being only 16.0%. According to the same resource, in 2015 16.7% of the transit buses in the U.S. were hybrid electric buses (HEB) which translates to about 12,000 HEBs in operation in the U.S. (APTA, 2016). But in recent years, completely zero emission buses are becoming progressively more popular and market ready. Plug in Battery Electric Bus (BEB) and Hydrogen Fuel Cell Bus (FCB) are both on-site zero emission options available for transit agencies. Technology related to battery electric buses have rapidly matured, making it one of the most desirable replacements to traditional combustion engine buses. A major reason apart from it being zero emission (on-site) is the recent improvement in cost competitiveness of these options. Other zero emission technologies are yet to become viable replacements in terms of cost. Since Foothill Transit in California started the journey with testing BEBs in 2010, many transit agencies have been considering migrating to an all-electric fleet. To date, more than 60 transit agencies in the US have either already taken up a test fleet or plan to deploy electric buses. Nevertheless, the common belief among public agencies remain that electric is equivalent to expensive. Although many of the agencies have performed an economic analysis of BEB adoption, there exists little to no study regarding the optimization of the fleet conversion.

Replacement studies are scarce in public transportation field. Most HEB or BEB fleet adoption models focus on purchasing process rather than replacement process. When switching the entire fleet or most part of it to a different technology, transit agencies must consider two important tradeoffs; Cost of owning and operating the buses and emission produced by the buses. These costs are function of market condition and technology development and so, can be assumed to vary over the years. Moreover, the cost of operation and maintenance (O&M) increases with the age of the bus. So, replacing older buses comes with higher capital cost but lower O&M cost (Feng and Figliozzi, 2012). These costs also vary across powertrain type of the buses.

The overarching objective of this paper is to help transit agencies make more informed decision regarding alternative technology adoption into an existing fleet. This paper is to finds an optimum bus replacement strategy and provides an optimum fleet mix consisting of hybrid electric and battery electric vehicles including the required charging infrastructure under budget and demand constraints using data from the Connecticut Department of Transportation (CTDOT). The model used in this paper is transferrable to other transit agencies wishing to find an optimum replacement schedule for any type of fleet mix under any budget or demand limitations. The model aims to minimize the life cycle cost of owning and operating a fleet mix along with GHG emissions, taking into account economic factors, environmental factors, characteristics of vehicles, demand and initial fleet mix.

Section 1 of this paper includes an introduction and the motivation behind this research. It also summarizes the objectives of this paper. Section 2 includes the literature review. Section 3 encompasses of the methodology of this optimization study. Section 4 comprises of the model formulation. Section 5 includes the description of the case study and section 6 includes the scenarios that was run for that case study. Section 7 comprises of the results and explanation of the base case and several sensitivity analysis of optimization model. Finally, section 8 includes some discussions and concluding remarks for this paper.

2. Literature review

2.1. Bus fleet cost analysis

Life cycle cost (LCC) is an estimate of the total purchasing, operating, maintenance and salvage cost of an alternative over the life span. LCC of diesel, hybrid and electric vehicles have been studied extensively (Xu et al., 2015, Delucchi and Lipman, 2010). Electric vehicles are usually associated with high purchasing cost but lower operating and maintenance cost. In most studies comparing buses of different fuel technologies, the cost of owning and operating a single vehicle of each type is compared. This study compares the LCC of the entire vehicle fleet instead of a single vehicle. Emission studies are also abundant comparing GHG emission of alternate fuel vehicles. In most of the studies, electric vehicles have proven to be superior in terms of emissions and cost to others (Ou et al., 2010; Lajunen, 2014). Studies including both life cycle cost and emission consideration of *transit buses* are very rare (Lajunen, 2014). Moreover, most of these studies are comparison based, meaning, the effect of replacing fleet of buses have not been studied thoroughly. To elaborate, when Life cycle cost of a single diesel bus is compared with a single Electric bus, a variety of factors are ignored in the process. First, it is impractical to assume that a fleet of buses can be replaced by a fleet of alternative technology bus overnight. In this case the operation and maintenance (O&M) cost of a fleet of mixed technology buses is ignored. Next, the age of a bus, which plays an important role in the O&M cost and emission is ignored or simplified. Finally, the cost of operating and maintaining the required infrastructure to support an alternative technology buses also varies depending on how many buses are being introduced to an existing fleet. So, it is important to study the LCC of replacing a fleet of bus.

2.2. Optimizing replacement problems

Equipment replacement or product replacement studies have been performed comprehensively in the fields of operations research (Eilon et al. 1966), Industrial Engineering (Bellman, 1955) and management science (Meyer, 1971). A parallel replacement problem is considered when multiple assets are required to be replaced in a finite time horizon. In this context, assets themselves are independent of each other (Parthanadee et al., 2012) as they satisfy a common demand, consume a common budget and have similar costs associated with them. Depending on the types of product in the fleet in consideration, replacement problems can either be homogeneous or heterogeneous. Homogenous models assume that the products of same age and type must be replaced together whereas Heterogeneous models don't. Heterogeneous models are more appropriate when solving real-world fleet replacements where the budget constraints are of concern. These models generally apply integer programming (IP) (Simms et al., 1984, Hartman, 2000). This section includes a heterogeneous model which considers three fuel types of buses that, although serve the same demand and utilize common budgets, have different cost structures associated with them. The basic dynamic framework developed by Hartman (2000) and Feng and Figliozzi (2012) is an IP problem though a dynamic framework was adopted in this study. Both of these studies utilize discrete time replacement, which means an unknown lifetime of the product. This study uses finite time horizon optimization which is more complex but is more practical when considering federal or state planning procedures and that transit vehicles have a documented expected lifetime.

Two relevant studies were executed for commercial electric trucks (Feng and Figliozzi 2012) and personal vehicles (He et al., 2017; Feng and Figliozzi, 2011; Miguel et al., 2011) which investigated the phasing out of conventional diesel vehicles and replacing them with electric but did not consider life-cycle cost in the model. Parthanadee et al. also studied the parallel replacement with alternative fuel consideration for vehicle fleet but that study was for personal automobile and it didn't include cost related to transit bus infrastructure. Feng and Figliozzi (2012) conducted one of the most extensive studies with real-world data considering both LCC and emission factors of hybrid and diesel buses to find an optimum replacement scheduling. That study, however only incorporated diesel and hybrid electric buses, excluding more recent technologies such as battery electric and hydrogen fuel cell buses.

In this study, a cumulative ownership, operation, maintenance, and emission cost over the design period was used to calculate the LCC of the fleet instead of the more conventional analysis method which is performed at a certain point of a product's life (Riechi et al., 2017). This method yields an optimum replacement timing cycle and a corresponding equivalent annual cost. Moreover, this study incorporates a more detailed GHG emission calculation instead of a per mile-based calculation used in prior work (e.g., Feng and Figliozzi, 2012). This study also includes a number of parameters that were not considered in the aforementioned studies such as costs for owning, operating and maintaining supporting infrastructures, learning cost, maximum expected GHG levels, maximum and minimum age of buses in fleet etc. This study also finds an optimized BEB adoption percentage by relaxing

To summarize, this study aims to find the optimum schedule for replacing diesel transit buses with a mix of HEB and BEB for DOT, considering the cost of building and maintaining charging infrastructure for BEB. Also, the dynamic nature of the transit system is considered where the demand for transit changes due to either population increase or changes in travel behavior. Mixed-Integer Programming (MIP) is used to model the fleet replacement problem as described above.

3. Methodology

For this study, a mixed integer program is formulated to find the optimum replacement schedule for buses. The model is deterministic in nature which means that all the input values must be known a priori. Fig. 1 shows the inputs, constraints, and outputs of the model.

The input values include economic factors, vehicle characteristics, environmental factors and available resources. The economic factors include annual budget, fuel prices, discount rate, rate of price increase, and the social cost of CO_2 SCC which is as a measure of economic impact of GHG emission. The environmental factors include emission factors and global warming potentials (GWP) for different GHGs. Vehicle-specific factors include capital costs and O&M costs of different bus types and infrastructure, salvage value, fuel economy, and maximum life span. Demand is the annual requirement of fleet size considering changes in transit ridership and population changes. Lastly, the initial fleet mix includes the age of all current vehicles in the fleet. The cost function has two major components consisting of life cycle cost and GHG emission cost for the entire fleet over the life span of the project.

The output of the model consists of number of bus purchased in a year, number of bus salvaged in a year, charging infrastructure built, existing number of buses and infrastructure, and cost breakdown in a year. The model was formulated in GAMS and solved using CPLEX.

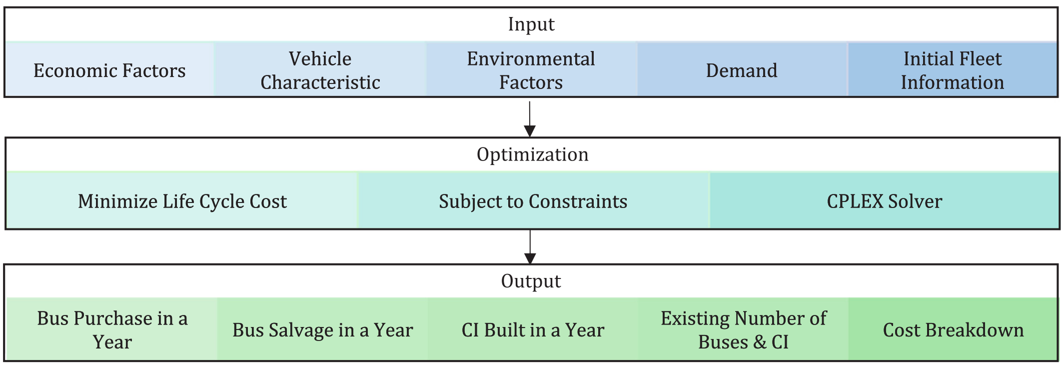

Fig. 1. Model flow.

4. Model formulation

The framework for the mathematical model for a fleet replacement used in this study is similar to Feng and Figliozzi (2012). The primary difference is the finite time structure nature of this study. As discussed earlier, another major difference is that this study includes a number of additional parameters and also aims to meet the target GHG level at the end of the analysis period.

4.1. Indices

$i \in \{0, 1, 2\cdots, I\}$	Age of vehicle
$k \in \{1, 2, ..K\}$	Types of fuel technology
$t \in \{0, 1, 2, ..T\}$	Number of years
$g \in \{1, 2, ..G\}$	Type of greenhouse gas

4.2. Decision variable

$P_{t,k}$	Number of *k* type vehicle purchased at year *t*
Q_t	Number of charging infrastructure installed at year *t*
$R_{t,k,i}$	Number of *i*-year old *k* type vehicle salvaged at year *t*
$A_{t,k,i}$	Number of *i*-year old *k* type vehicle in operation at year *t*
C_t	Number of charging infrastructure in operation at year *t*

4.3. Parameters

B	Total annual budget
$e_{t,k}$	Capital cost of *k* type vehicle at year *t* (bus cost + warranty cost)
$f_{k,i}$	Operation and maintenance cost per mile for *i*-year old *k* type vehicle
j_k	Fuel cost per mile for *k*-type vehicle
l	Capital cost of charging infrastructure
n	O&M cost of charging infrastructure
s_k	Salvage value of *k*-type bus
δ_t	Social cost of CO_2 at year *t*
$\pi_{t,k}$	Renewable fuel percentage of *k*-type bus in year *t*
γ_t	Expected GHG level at year *t*
$\theta_{k,g}$	Emission factor for *g*-type GHG by k type vehicle (kg/mile)
ϑ_g	Emission factor for *g*-type GHG (kg/mile)
d_t	Demand of vehicle at year *t*
m	Average annual mileage in year *t*
α	Discount rate
β	Rate of price increase
φ	Maximum age of a bus in operation
ρ	Minimum age of a bus in operation
ω	Minimum percentage of Battery Electric bus at the end of the analysis period
$h_{k,i}$	Number of *k* type *i* year old vehicle available at time 0 ***(Initial Condition)***

4.4. Objective function

$$\min\ z = \sum_{t=0}^{T} LCC_t = \sum_{t=0}^{T} \left[\begin{array}{c} \sum_{k=0}^{K} P_{t,k} e_{t,k} + \sum_{k=0}^{K} \sum_{i=1}^{I} (A_{t,k,i} f_{k,i} + A_{t,k,i} j_k) m \\ + Q_t l + C_t n - \sum_{k=0}^{K} \sum_{i=1}^{I} R_{t,k,i} s \end{array} \right] * \frac{(1+\alpha)^t}{(1+\beta)^t}$$
$$+ \sum_{k=1}^{K} \sum_{t=0}^{T} \sum_{i=1}^{I} \sum_{g=1}^{G} \frac{A_{k,t,i} * \theta_{k,g} * \vartheta_g * \delta_t * m * (1-\pi_{k,t})}{1000} \tag{1}$$

4.5. Constraints

$$\sum_{k=1}^{K} (P_{t,k} * e_{t,k} + Q_t * l) \le B_t \qquad \forall\ t\cdots \tag{2}$$

$$\sum_{k=0}^{K}\sum_{i=1}^{I}A_{t,k,i} \geq d_t \quad \forall\ t\cdots \tag{3}$$

$$R_{0,k,i} + A_{0,k,i} + P_{0,k} = h_{k,i} \quad \forall\ k,\ t\cdots \tag{4}$$

$$P_{t,k} = A_{t,k,0} \quad \forall\ k,\ t\cdots \tag{5}$$

$$A_{t,k,i} = A_{(t-1),k,(i-1)} - R_{t,k,i} \quad \forall\ k,\ t,\ i\cdots \tag{6}$$

$$R_{t,k,i} = 0 \quad \forall\ t,\ i \in \{0,\ 1,\ \cdots\rho\}\cdots \tag{7}$$

$$A_{t,k,i} = 0 \quad \forall\ t,\ i \in \{\varphi,\ \varphi+1,\ ..I\}\cdots \tag{8}$$

$$P_{t,1} = 0 \quad \forall\ k,\ t\cdots \tag{9}$$

$$C_t \geq \sum_{i=1}^{I}A_{t,K,i} \quad \forall\ t\cdots \tag{10}$$

$$C_0 = Q_0 \quad \forall\ t\cdots \tag{11}$$

$$C_t = C_{(t-1)} + Q_t \quad \forall\ t\cdots \tag{12}$$

$$\sum_{k=1}^{K}\sum_{i=1}^{I}\sum_{g=1}^{G}(\theta_{k,g} mA_{t,k,i} * \vartheta_g * (1-\pi_{t,k})) \leq \gamma_t \quad \forall\ t \tag{13}$$

$$\sum_{i=0}^{I}A_{T,k,i} = \omega * \sum_{k=1}^{K}\sum_{i=0}^{I}A_{T,k,i}\cdots \tag{14}$$

$$P_{t,k},\ R_{t,k,i},\ A_{t,k,i},\ Q_t,\ C_t \in \mathbb{Z}^{+}\cdots \tag{15}$$

The objective function, Eq. (1), minimizes the sum of purchasing, operating and maintaining cost of the entire fleet of buses including charging infrastructure, fuel cost, salvage value, and emission costs.

Eq. (2) ensures that annual expenditures do not exceed the available annual budget, based on the CTDOT published budget for the next 5 fiscal years (CTDOT, 2017). The study assumes an average capital budget each year, though the model could accommodate distinct annual budget amounts if those numbers were known with certainty. Expression (3) ensures that total number of buses in operation at any given year satisfies the demand. Expression (4) establishes the initial fleet mix. Expression (5) satisfies the purchase new vehicle only rule (PNOR). Which mathematically translates to – at any given year, the number of any type of bus purchased sets the value of zero-year-old buses. Expression (6) increments the age of a bus by one year at the end of every year. Expression (7) ensures that the buses must be in operation for at least 12 years before than can be salvaged, in accordance with state and federal requirements (FTA, 2007). Expression (8) requires that when a bus reaches its maximum age, it cannot be in operation anymore. Federal law mandates that vehicles at the end of their service life must be salvaged due to safety and environmental hazard (FTA, 2007). For Connecticut, the state of practice is 15 years, based on data describing the existing fleet. This also satisfies the older vehicle selling rule (OVSR) which is adopted from an optimal rule called the "older cluster replacement rule" (OCRR), which states that a machine of age *i* is replaced only if all machines of greater age have been replaced (Hopp et al., 1993). Expression (9) indicates that purchase of diesel buses is restricted for this study. As mentioned earlier, a major aim of this study is to reduce GHG emissions in a cost-effective manner. As replacing diesel buses does not accomplish this goal, the purchasing of new diesel buses is disallowed. Expression (10) satisfies the required number of charging infrastructure at any given year for the battery electric bus fleet. According to current practices in battery electric buses, the transit agency must have one depot charger to charge each bus. Expression (11) specifies the initial condition for charging infrastructure. Expression (12) ensures that charging infrastructures remains in use in the following years until the design year is reached. Also, newer stations are in operation immediately after construction. Expression (13) ensures that the GHG emission from the existing bus fleet at any given year is less than the emission goal set by CTDOT. Emission targets in each year is projected linearly from the emission goal set at the end of analysis period. Expression (14) ensures that at least some percentage of the total fleet at the end of design year is battery electric (or any other technology of interest). This may take any value from zero to one hundred. Expression (15) satisfies the MIP structure that says all the decision variables must be non-negative integers.

5. Case study

In 2008, the Connecticut Global Warming Solutions Act set mandatory targets for reducing GHG to at least 10% below 1990 levels by 2020 and 80% below 2001 levels by 2050 (CT, 2012). On Earth Day 2015, the Governor's Council on Climate Change, otherwise known as the GC3 was created for the sole purpose of examining the effectiveness of existing policies and regulations required to reduce GHG emission in the state. Connecticut Department of Transportation (CTDOT) has requested Connecticut Academy of Science and Engineering (CASE) to take up a study to identify strategies to achieve a vision of pathway to minimize their carbon footprint (CASE, 2018). As a result of that study, it was concluded that DOT should replace their diesel bus fleets with alternate fuel

Table 1
2017 Bus fleet composition for CTDOT.

Age	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
DB	0	0	0	11	0	0	0	58	37	2	2	65	0	48	84	62	41	23	0
HEB	0	0	0	3	81	0	8	10	14	0	0	0	0	0	0	1	0	0	0

technology buses to achieve their emissions reduction goals.

The bus fleet owned and operated by CTDOT was used as the base for a case study for this model. First, background data was collected from all the agencies that operate under CTDOT for the emission calculations and economic estimates. These included 2016 data for fleet composition (Diesel/Hybrid), vehicle age, fuel usage per mile, fuel efficiency per mile), and vehicle miles travelled per bus (VMT). The temporal data was then averaged over a 12-month period to get average fuel usage, fuel efficiency and VMT. Table 1 shows the age and number of Diesel Buses (DB) and Hybrid Electric Buses (HEB) in operation for CTDOT in 2017. This was used as the initial condition of the data (h_{ki}).

Purchasing, operating and maintaining costs; fuel economy, fuel cost inputs were collected from either FuelCost2 (TCRP, 2011) or from NREL reports (). The GHG emission factors and GWPs were collected from EPA's simplified calculator (EPA, 2017a, 2017b) and GREET 2017. The base year of the finite analysis period is 2018 and end year is 2030. These assumptions are listed in Table 2.

6. Scenarios

The base case in this study refers to the scenario where the following assumptions are made.

1. Total bus fleet is replaced completely with battery electric buses (ω = 100%) by 2030.
2. GHG level in 2030 is expected to be 35% below GHG level in the base year.
3. A declination in cost of battery electric bus is 12.5% in 2030 from 2018.
4. Social cost of carbon (SCC) is assumed to be \$36/Metric Ton of CO_2 equivalent (MT CO_2 e) (EPA, 2016).
5. Discount rate and rate of price increase are both assumed to be 3%.
6. Renewable electricity (zero carbon) is assumed as 30% of the total state energy portfolio in the year 2030.
7. Transit ridership remains constant throughout the analysis period.

The results for the base case are discussed in the results section. The subsequent part of this study involves sensitivity analysis of multiple scenarios consisting of different combinations of the above-mentioned parameters. First, BEB technology adoption percentage at the end of analysis period is varied from 45% to 100%. Increase in percentage adoption of BEB means further reduction in GHG emission. Next, this model was tested for the sensitivity to different levels of BEB adoption. Cost declination of BEBs are used from 0% to 25% of that of base year. Literature suggests that increase in BEB market share and the decline in battery pack costs will result in declination of BEB purchase costs significantly in the coming years (McKinsey and Company, 2017). This is consistent with the reduction in electric automobiles costs over the past 7 years. Next, expected GHG level in 2030 is varied from 15% to 95% of that of the base year. As there is no hard GHG level target set for 2030, the model was tested for a several different scenarios. Next, SCC is varied from \$36 to \$350/MT CO_2 e. The base social cost of carbon (\$36) takes into account economic damage due to climate change. But there has been a variety of studies showing different levels of SCC. One study suggests an unweighted value of \$51.4/ MT CO_2 e and a value of \$329/MT CO_2 e weighted for equity (Anthoff et al., 2011). If temperature change is taken into account then under a business-as-usual emissions scenario, an average SCC value of \$96/ MT CO_2 e can be assumed (Hanemann, 2008). Ackerman and Stanton (2012) reported values between \$241 and \$445/ MT CO_2 e. Renewable electricity is varied from 20% to 30% in 2030. The

Table 2
Economic parameter assumptions.

Input data		DB	HEB	BEB	FCB
Estimated capital bus cost	2018	\$450 K[a]	\$600 K[a]	\$800 K[b]	\$1.356 M[c]
	2030	\$450 K	\$600 K	\$700 K[c]	\$1.356 M
Fuel price		\$2.50/gal[d]	\$2.50/gal[d]	\$0.12/kwh[d]	\$7.81/DGE[d]
Fuel cost (\$/mile) (calculated)		\$0.68/mile	\$0.49	\$0.25	\$1.12
Bus maintenance cost (\$/mile)		\$0.45[a]	\$0.47[a]	\$0.16[e]	\$0.21[b]
Fueling infrastructure capital cost		–	–	\$50 K/bus[e]	\$100 k/bus[f]
Fueling infrastructure O&M cost (Annual)		\$189/bus[a]	\$163/bus[a]	\$38/bus[b]	\$140 K/unit[b]
Learning cost multiplier	Year 1	–	–	1.1[d]	1.1[d]
	Year 2	–	–	1.2[d]	1.2[d]
Annual mileage of bus (miles) – 36,000[g]					
Annual discount rate – 3% (assumed)					
Annual rate of price increase – 3% (assumed)					

(a): CASE (2018); (b) ARB (2016); (c) Eudy and Post (2017); (d): SAIC (2011); (e): Eudy et al. (2016); (f): Melaina and Penev (2013)

Table 3
Purchasing schedule for base case.

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Purchasing schedule: buses													
HEB	27	0	0	0	0	0	0	0	0	0	0	0	0
BEB	100	100	100	2	52	68	8	1	81	14	0	1	27
Annual capital cost (Million $)													
	113.8	93.6	92.7	3.6	47.1	62.4	6.0	0.74	72.0	11.9	0	0.71	24.3
Installation schedule: charging infrastructure													
	9	8	8	1	4	6	0	0	7	1	0	0	3

2017 level of renewable electricity in the state of Connecticut is about 17% of the total electricity (DEEP, 2016). This is expected to rise over the course of coming years to reach the expected level of 85% by 2050 (DEEP, 2017).

Finally, the public transportation ridership is assumed to increase up to 6% (compared to current 3%) by 2030. Fleet size and therefore total bus VMT and fuel usage are a function of transit ridership. This assumption is consistent with the goals established in programs such as 'Let's Go CT!' which evaluates and seeks to improve the statewide bus transit system (CTDOT, 2015). The 30-year plan includes expansion of bus service by 25%, providing residents in urbanized areas access to bus within half-mile, integrating real time information, extending CTFastrak to increase access to jobs and education, targeting University students with U-pass etc. (CTDOT, 2015). CTFastrak alone has seen 23% increase in total corridor passenger trips in July 2016 compared to that of July 2015. Looking at the data of 33.25 million passenger trips in 2006 and 42.16 million in 2014 and taking into account state's effort to become more sustainable, it can be assumed that the ridership will keep increasing in the coming years. An appropriate fleet turnover strategy is required to meet the increased demand.

7. Results

7.1. Base case

For the base case, the results of optimized purchasing schedule is shown in Table 3. The total life cycle cost for the base case is $638 million and the total GHG emissions are 0.198million MT CO_2equivalent. Total allocated capital budget is not utilized completely in all the years. The first three years are the ones with highest capital cost associated with them.

Table 4 shows the salvaging schedule for the base case. The initial fleet composition included many older diesel buses. From the salvaging schedule it can be seen that the model is disposing of them due to high cost associated with operating and maintenance of older buses and the maximum age constraint. These are being replaced by battery electric buses in all years of the analysis period with the exception of the initial year where a number of hybrid buses are being purchased due to the available capital budget.

Fig. 2 is a graphical representation of the table that shows a gradual declination in number of diesel and hybrid buses and increase in BEBs as the final year of analysis period is approached.

7.2. BEB adoption percentage

For the base case, it was assumed that the total fleet will be replaced completely by low emission battery electric buses by 2030. Obviously, a higher percentage of BEB will result in lower emission and hence lower carbon cost. But the high purchase cost associated with BEBs may lead to the belief that this is not the most cost-effective solution. The model was rerun for different percentage of BEBs in operation at the end of analysis period to find a least cost solution. In other words, the ω value in expression

Table 4
Salvaging schedule for base case.

Age	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
DB													
12	0	0	2	2	37	58	0	0	0	11	0	0	0
13	0	0	65	0	0	0	0	0	0	0	0	0	0
14	0	15	0	0	0	0	0	0	0	0	0	0	0
15	62	84	33	0	0	0	0	0	0	0	0	0	0
16	41	0	0	0	0	0	0	0	0	0	0	0	0
17	23	0	0	0	0	0	0	0	0	0	0	0	0
HEB													
12	0	0	0	0	14	10	8	0	81	3	0	0	27
15	1	0	0	0	0	0	0	0	0	0	0	0	0
Total													
	127	99	100	2	51	68	8	0	81	14	0	0	27

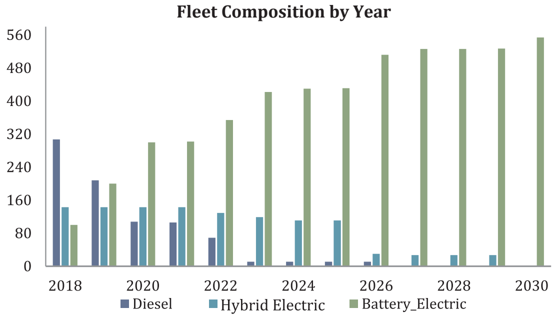

Fig. 2. Fleet composition in each year of analysis period – base case.

(14) was given a value of 0.0 – 1.0 to get the sensitivity of the model to BEB adoption percentage.

$$\sum_{i=0}^{I} A_{T,k,i} = \omega * \sum_{k=1}^{K} \sum_{i=0}^{I} A_{T,k,i}$$

Fig. 3 shows the variation in life cycle cost and carbon cost for different percentage of BEB adoption.

The figure shows very interesting results. Although the relationship between emission and BEB percent adoption remains intuitive, the costs fluctuate. This is due to the fact that the Eq. (14) is making this model is very restrictive. In order to meet this constraint, buses are either being purchased or salvaged when it may not be necessary which results in higher costs even with lower adoption. The change in the color of represents the different percentage of BEB adoption. The darkest blue represents 100% BEB adoption at the end of the analysis period and the lightest shade is 45%. Below 45% adoption of BEBs, the model become infeasible due to the emission constraint.

From the figure one can infer that at 80% BEB adoption by 2030, the LCC is the lowest for this case study ($617.4 Million). This is 3.2% lower than the base cost which represents 100% BEB adoption. But this 20.5 million reduction in cost comes at a price of 19.25% increase is greenhouse gas emissions over the 12 year period. In other words, in this scenario, a reduction in cost of $537 is encountered for *not reducing* each Metric ton of CO_2 equivalent.

This constraint (Eq. (14)) can be relaxed by modifying the equation as follows, allowing less than a certain percentage of replacement of buses.

$$\sum_{i=0}^{I} A_{T,k,i} = \omega * \sum_{k=1}^{K} \sum_{i=0}^{I} A_{T,k,i} \tag{16}$$

With this modified constraint, the model finds the most cost effective BEB fleet mix using an inequality. When the value of ω is varied, in this case, for any value above 80% adoption (0.8–1.0), the model gives the same result. The model yields a LCC of $616.6 Million which is lower than any percentage used in the restricted model. If the values are varied further, the cost start increasing. This means that the LCC of $616.6 million is the least cost solution for these set of constraints and the adoption schedule for this scenario is the optimized percentage of BEB adoption. Fig. 4 shows the effect of percent BEB adoption on LCC and GHG Emission due to relaxed

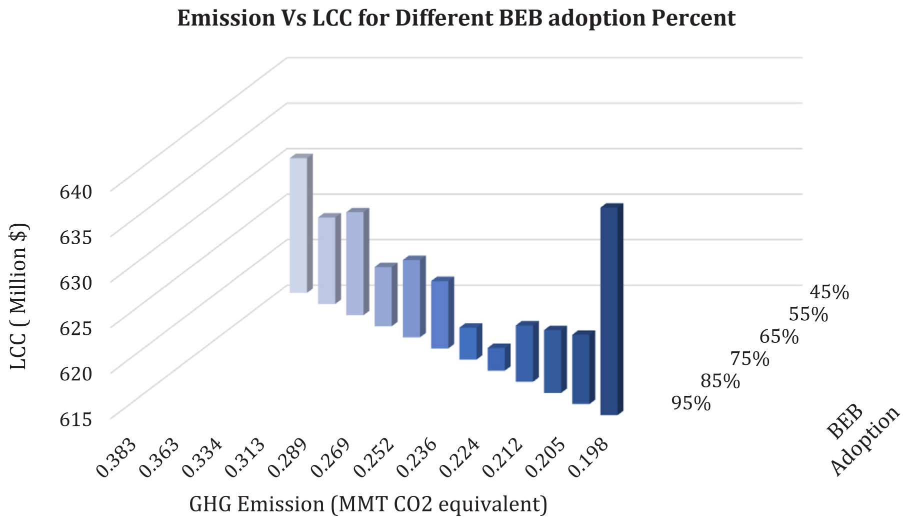

Fig. 3. Effect of percent BEB adoption on life cycle cost and emission due with restricted constraint.

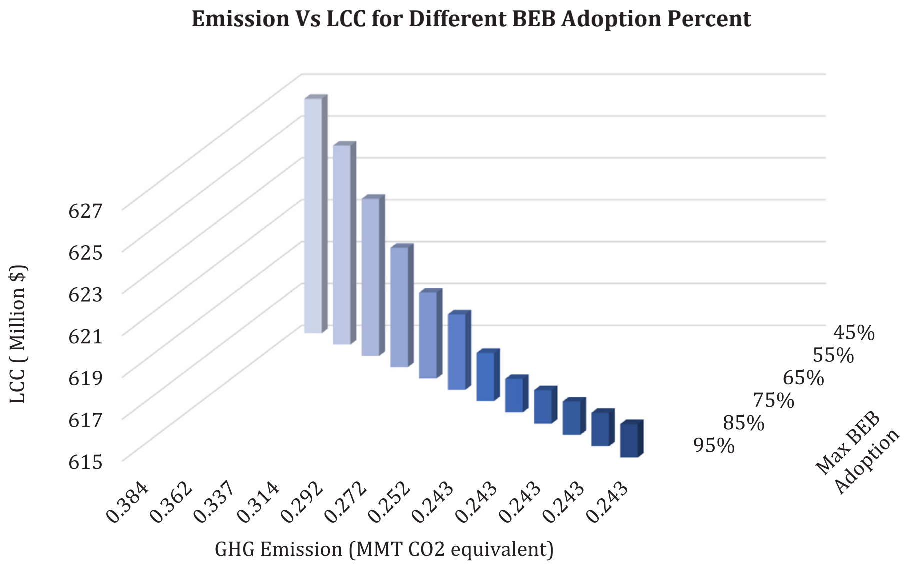

Fig. 4. Effect of percent BEB adoption on life cycle cost and GHG emission due to relaxed constraint.

constraint.

The least cost solution of \$616.6 million occurs at 79% BEB adoption. This result shows that it possible to save \$21.3 million by retaining 21% of the fleet as Hybrid Electric. Fig. 4 shows the effect of fleet composition with this more relaxed constraint (Eq. (16)). This constraint also eliminates the general hypothesis that higher percentage of BEB will result in higher cost. For example, the cost of replacing 79% vehicles with battery electric buses is 4.5 million dollars lower than 60%. But this 79% replacement comes at a cost of 21.9% increase in GHG emission than the base case.

Table 5 shows the replacement schedule, cost and required charging infrastructure for the least cost solution. The results are very similar to the base case except the later few years. This is because, by the time the analysis reaches 2025, the fleet composition had already achieved the emission goals at the end. So, despite the fact that HEBs produce more emission, the model allows the purchase to minimize the initial capital cost.

Fig. 5 shows the graphical representation of existing buses in the analysis period with 79% BEB adoption.

7.3. BEB cost decline

Although declination the cost of battery pack for electric vehicle, and hence the declination of overall capital cost of BEBs is supported by several recent studies (McKinsey and Company, 2017), a sensitivity analysis was conducted for the pessimistic case in which BEB capital cost remains constant. With no decrease in the cost of BEBs, the base case replacement schedule doesn't hold. The life cycle cost for the optimized replacement schedule is \$654.7 million which is about 2.6% higher than the base case. But the replacement schedule remains the same as the base case.

For the relaxed constraint (Expression (16)), it can be found that the least cost solution occurs at 65% BEB adoption for no decline in BEB cost over the analysis period. The cost is 1.2% higher than that of 79% BEB adoption in the baseline BEB cost decline scenario. But GHG level at this replacement schedule is also 20.9% higher than that at 0.293 MMT of CO_2 equivalent. Table 6 shows the purchasing schedule for no declination in BEB cost for 65% BEB adoption. The purchasing schedule up to 2022 is identical to the 79% BEB adoption at base cost declination scenario. This is due to the fact that there are high number of older diesel buses in the DOT fleet in the earlier years of the analysis period which are required to be replaced due to the age constraint and also the GHG level

Table 5
Replacement schedule for 79% BEB adoption.

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
Purchasing schedule: buses													
HEB	27	0	0	0	0	0	0	0	80	14	0	1	0
BEB	100	100	100	2	52	68	8	1	1	0	0	0	0
Annual capital cost (Million \$)													
	113.8	93.6	92.7	3.35	47.1	62.4	6.0	0.74	52.7	9.1	0	0.65	0
Installation schedule: charging infrastructure													
	9	8	8	1	4	6	0	0	0	0	0	0	0

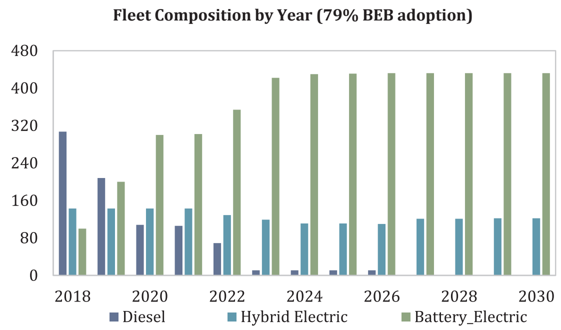

Fig. 5. Fleet composition in each year of analysis period – optimized percent (79%) of BEB Adoption.

Table 6
Purchasing schedule for variation in decline in BEB cost.

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030
No decline in cost: purchasing schedule of buses (65% adoption)													
HEB	27	0	0	0	0	65	5	1	81	14	0	1	0
BEB	100	100	100	2	52	3	3	0	0	0	0	0	0
25% decline in cost: purchasing schedule of buses (96% adoption)													
HEB	27	0	0	0	0	0	0	0	0	0	0	0	0
BEB	100	100	100	2	52	68	8	1	81	14	0	1	0

constraints. When the minimum GHG levels are satisfied, hybrid bus purchase prevails as the best choice over battery electric buses due to lower purchasing cost.

For a 25% reduction in BEB capital cost by 2030, the cost becomes \$600,000. Although the replacement schedule remains the same as the base case, the life cycle cost becomes \$649.3 million resulting in a 4.9% decrease in cost. But for the relaxed constraint, the least cost solution occurs at 95% BEB adoption. In that case, LCC becomes \$632.5 million resulting in a 1.5% decrease in cost compared to 78% adoption of BEB with base cost decline scenario. In this scenario, the emission is also 17% less than the former scenario at 0.2004 million metric tons of CO_2 equivalent. Table 6 shows the schedule for both 0% decrease and 25% decrease of BEB purchase cost for the relaxed constraint scenario.

7.4. Social cost of carbon

Changes in the social cost of carbon has an impact on life cycle cost and carbon cost. For higher SCC, the life cycle cost increases significantly. Table 7 shows the results for a few different SCC.

The overall replacement schedule remains the same for the more restricted constraint (expression (14)) regardless of the SCC. In this scenario, the only portion of this particular model that is sensitive to SCC changes is the carbon cost. This may not hold true for any other set of initial condition inputs. For the relaxed constraint, when higher SCC values are considered compared to the base case of \$36/Metric Ton CO_2, BEB adoption percentage changes to 96% and follows the replacement schedule as 25% cost decline in BEB (Table 5). This is because, with the higher social cost of carbon, diesel and hybrid buses are now costlier to operate. A sensitivity analysis for zero dollars SCC was also considered. The replacement schedule unsurprisingly remains same for the restricted constraint (expression (14)). But an interesting result is found when the model is run for relaxed constraint (expression (16)). Even without considering any SCC value, the model results in 78% BEB adoption being the most optimal solution for this problem.

Table 7
Sensitivity to social cost of carbon.

SCC in 2030 (\$/Ton CO_2)	BEB Adoption (%)	Life cycle cost (Million \$)	GHG level (MMT CO_2 e)
Restricted model			
0	100	658.2	0.1980
125	100	685.4	0.1980
360	100	734.3	0.1980
Relaxed model			
0	78	636.9	0.2424
125	95	666.3	0.2006
360	95	715.9	0.2004

Table 8
Sensitivity to change in GHG level in 2030.

Reduction of GHG level in 2030 (%)	BEB adoption (%)	Life cycle cost ($)	GHG level (MMT CO_2 e)
0–75	79	645.9	0.2419
85	88	646.2	0.2164
95	97	654.6	0.1995

7.5. GHG level in 2030

The base case assumption starts enforces a 35% reduction in GHG level by 2030. A sensitivity analysis was done for a few other cases by lowering and increasing the reduction in GHG levels by 2030. Table 8 shows the results of the analysis.

For reduction in GHG levels up to 65% by 2030 of 2018 level results in the same replacement schedule as the base case. But as the reduction level increases, optimum level of BEB adoption changes. At 95%, the optimized BEB adoption is maximum at 98%. The replacement schedules are identical to the optimized BEB adoption up to year 2025 but varies in the coming years.

7.6. Transit ridership in 2030

Transit Ridership in 2030 is expected to increase in some statewide planning scenarios. An algorithm was used to estimate the fleet size requirements for various ridership scenarios (Islam, 2018). A sensitivity analysis for varying levels of transit ridership is shown in the following table (see Table 9).

From the table, it can be seen that with the increase of transit ridership, optimum BEB adoption percentage decreases. This may be due to budget constraint. This may be due to budget constraint.

7.7. Sensitivity analysis summary

The purpose of sensitivity analysis was to find the responsiveness of the model to different constraints and parameters. Optimum solutions under different conditions can be used to aid replacement decisions. A manual calculation was performed to find the life-cycle cost including age and emissions with the DOT schedule discussed in chapter 3. Total LCC, in this case, was $867.5 million and GHG emission was found to be 0.313 MMT CO_2 equivalent. Comparisons were drawn with this scenario based on cost saved by adopting different optimized schedules. Summary of some of the optimized schedule is shown in the following table (Table 10).

The first two columns of the table represent the different sensitivity scenarios and BEB adoption in those scenarios. The third column shows the LCC cost of different scenarios and BEB adoption percentages. The fourth column is the GHG emission in each scenario. The fifth column represents the decrease in cost compared to the DOT Schedule. The final column shows % of emission reduction by adopting different optimized schedule compared to DOT schedule. For example, in restricted BEB adoption percentage case, about $22.4 million is being saved and about 36.7% of emission is being reduced. Whereas, for the relaxed schedule, cost reduction is $41.6 million but emission reduction is only 22.7%.

Fig. 6 shows a graphical representation of emission and life cycle cost from unoptimized and optimized replacement schedules. It is clear from Fig. 6 that in order to gain the maximum emission and cost reduction, the transit agency of concern must optimize their fleet replacement schedule. This will not only allow them to achieve their emission goals, but also minimize financial burden.

8. Discussion

8.1. Policy implications

In order to maintain the health of our cities and the well-being of the environment around us, embracing low emission technologies is essential. But how and when a policy induces the adoption to these are imperative in order to protect the users and service

Table 9
Sensitivity to change in transit ridership in 2030.

Transit ridership (%)	BEB adoption (%)	Life cycle cost ($ Million)	GHG level (MMT CO_2 e)
Restricted model			
4	100	744.7	0.2010
5	100	823.2	0.2039
6	100	906.5	0.2118
Relaxed model			
4	75	723.1	0.2548
5	74	800.1	0.2669
6	71	877.7	0.2846

Table 10
Sensitivity summary.

Case	BEB adoption	LCC ($ Million)	GHG emission (MMT CO2e)	Cost reduction ($ Million)	% emission reduction
DOT replacement schedule					
	100%	687.5	0.3130	–	
% BEB adoption in 2030					
Restricted	100%	665.1	0.1980	22.4	36.7%
Relaxed	78%	645.9	0.2419	41.6	22.7%
BEB cost decline by 2030 compared to 2018					
No decline	100%	682.9	0.1980	4.6	36.7%
	76%	653.2	0.2474	34.3	21.0%
25% decline	100%	649.3	0.1980	38.2	36.7%
	95%	623.5	0.2004	64.0	36.0%

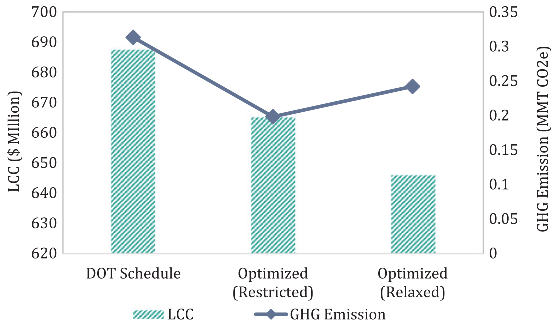

Fig. 6. Comparison of DOT and optimized bus replacement schedule.

providers from technological and economic shocks. This research showcases bus fleet replacement strategies under different scenarios. It has the potential to make a significant contribution by helping the policy makers in their decision-making process.

When dealing with budget constraints and emission reduction goals, it is impossible to find the optimum solution without some sophisticated algorithm. This paper does just that. It is up to the policy makers to decide which constraints they anticipate and more importantly what they want to achieve. If lowest possible emission is the primary goal, then a transit agency may wish to adopt 100% BEBs within their analysis period which will result in higher costs. On the other hand, if the decision makers wish to gain cost efficiencies by maintaining a portion of the fleet as Hybrid Electric, this research can also help them achieve that. This approach may appear controversial to some, as the least possible emission will not be achieved within the analysis period. But the model will still produce replacement schedules that will help them achieve their set goals. Moreover, this mixed fleet has the potential or added resilience benefits should a power source be unavailable during storms or other natural disasters. This research can form the base of methodological calculations that can help transit agencies win federal grants and the confidence of the people. In conclusion, this research can be a great resource for policy makers wishing to achieve their emission goals within budget constraints.

8.2. Methodological contribution, limitations and future research

The parallel fleet replacement model discussed in this paper is deterministic and gives a quantifiable understanding of the importance of shifting towards alternative fuel technology. The model gives flexibility to the user to introduce input values and also to set the scenarios for the desired outcome. Transit agencies can decide what is most important to achieve, be it cost reduction or emission reduction or achieving a certain level of technology adoption, the model can provide the optimum solution to their problem. The model is an improvement over previous studies regarding optimization of transit fleet replacement as it includes finite time horizon instead of discrete time frame and incorporates the inclusion of fueling infrastructure demand and costs. Also, as a part of the case study, various scenarios including SCC variation and cost variations were considered in a manner that is more representative of current and future scenarios. To sum up, this research satisfies its primary objective of helping transit agencies make more informed decision regarding alternative technology adoption into their existing fleet.

A substantial number of parameter values were assumed based on reliable, published documentation. However, some parameters values had significant variation based on different sources. Parameters associated with newer technologies, are not available through any single source of input. Moreover, a considerable amount of uncertainty remains regarding future scenarios and parameter values. Although sensitivity analyses were performed in an effort to capture some of the nuances of uncertainty but studying all possible variations of all input values were out of the scope of the study. For future research, the study could be revisited when more reliable single source data is available. As an alternative way of incorporating uncertainty into the model, a stochastic model can be studied instead of deterministic one. A stochastic model gives the probability of a potential outcome by incorporating random variation of inputs. It gives the likelihood of an event occurring within a confidence interval. Although deterministic models are more useful when exact solution is desired, they ignore the randomness of input. A stochastic model could overcome some of the uncertainty and risk associated with parameters. As newer technologies become available, this research can be updated for optimizing inclusion of those technologies into fleet mix.

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